

Linear Algebra

Warm Up

Exercise 1. The characteristic polynomial of a 3×3 matrix A is $\lambda^3 + 2\lambda^2 - 15\lambda$.

- (a) What is $\text{Tr}(A)$? (b) Is A invertible? (c) Is A diagonalizable?

Exercise 2. For each of the following, either come up with an example, or prove that no example exists.

- (a) A 2×2 matrix over $\mathbb{R}$ with no real eigenvectors.
- (b) A 2×2 matrix over $\mathbb{R}$ with exactly 1 real eigenvector (up to scaling).
- (c) A 2×2 matrix over $\mathbb{C}$ with no complex eigenvectors.
- (d) A 2×2 matrix over $\mathbb{C}$ with exactly 1 complex eigenvector (up to scaling).

Exercise 3. The **Cayley-Hamilton Theorem** says that a matrix A always satisfies its own characteristic polynomial. What that means is that if you write out the characteristic polynomial $f(x)$, and plug in the matrix A (even though it's a matrix and not a number), you always get the 0 matrix: $f(A) = 0$.

- (a) Suppose A is diagonalizable matrix over $\mathbb{C}$. Prove the Cayley-Hamilton Theorem for A . (Hint: you can prove a matrix is the zero matrix by showing it acts by 0 on a basis.)
- (b) Look at your example from Exercise 2 (b). Verify that it satisfies the Cayley-Hamilton Theorem.
- (c) Find an example of an $n \times n$ matrix that satisfies a polynomial of degree less than n .

Exercise 4. (August 2014 Problem 3, parts (a)-(c))

- (a) Let V be a finite-dimensional vector space over $\mathbb{C}$, and let $T: V \rightarrow V$ be a linear transformation. Show that T has an eigenvector.
- (b) Give an example of a nonzero finite-dimensional vector space V over $\mathbb{R}$ and a linear transformation $T: V \rightarrow V$ such that T does not have an eigenvector.
- (c) Does a linear transformation of an infinite-dimensional vector space over $\mathbb{C}$ have to have an eigenvector? Either prove this is the case, or give an example of a linear transformation of an infinite-dimensional vector space with no eigenvector.

Exercise 5.

- (a) Does every monic degree 2 polynomial over $\mathbb{C}$ occur as the characteristic polynomial of a 2×2 matrix with complex entries?
- (b) Does every monic degree 2 polynomial over $\mathbb{R}$ occur as the characteristic polynomial of a 2×2 matrix with real entries?

Work Out

Exercise 6. (January 1983 Problem 1) Suppose V is a finite dimensional vector space over $\mathbb{C}$, and $T: V \rightarrow V$ is a linear operator. Suppose the subspace of V spanned by all eigenvectors of T is 1-dimensional. Prove that the minimal polynomial of T is $(x - \alpha)^n$, where $n = \dim V$ and α is some complex number.

Exercise 7. (January 2022 Problem 2, parts (a),(b),(d)) For each of the following statements, either prove it or provide a counterexample.

- (a) Every matrix of finite order in $\mathrm{GL}_n(\mathbb{C})$ is diagonalizable.
- (b) Every matrix of finite order in $\mathrm{GL}_n(\mathbb{Q})$ is diagonalizable.
- (d) Every matrix of finite order in $\mathrm{GL}_n(\overline{\mathbb{F}}_p)$ is diagonalizable. (We write $\overline{\mathbb{F}}_p$ for the algebraic closure of the finite field $\mathbb{F}_p$.)

Exercise 8. (January 2017 Problem 2) Let $n > 0$ be an integer. Let F be a field of characteristic zero, V a vector space over F of dimension n , and $T: V \rightarrow V$ an invertible F -linear map such that $T^{-1} = T$. Denote by W the vector space of linear transformations from V to V that commute with T . Find a formula for $\dim(W)$ in terms of n and the trace of T .

Exercise 9. (January 2025 Problem 4) Let V be a dimension 4 vector space over $\mathbb{C}$ and $\phi: V \rightarrow V$ an operator. Suppose that a dimension 2 subspace W of V is invariant in the sense that $\phi(W) \subset W$; then ϕ induces operators $W \rightarrow W$ and $V/W \rightarrow V/W$.

Suppose that the Jordan form of ϕ on both W and V/W is a single 2×2 Jordan block with eigenvalue 0. Find all possible Jordan forms of ϕ on V .

(For full credit, you need to provide examples that show your answers indeed can happen and prove that there are no other possibilities.)